Example 8.8

In the experiment of tossing an unbiased coin (or synonymously balanced or fair coin), the sample space is $S = \{H, T\}$. What is the probability of getting head or tail?

Solution :

If the events A_1 and A_2 are defined as $A_1 = \{H\}$ and $A_2 = \{T\}$, then $S = A_1 \cup A_2$. Here, the events A_1 and A_2 are mutually exclusive, because they cannot occur together. Hence, by using Axiom-3, it can be written as

$$P(S) = P(A_1)+P(A_2)$$

Since the number of elementary events in S in favour of the occurrence of A_1 and A_2 is one each, they have equal chances to occur. Hence, $P(A_1) = P(A_2)$. Substituting this, it follows that

$$1 = P(A_1)+P(A_2) = 2P(A_1)$$

It gives that $P(A_1) = \frac{1}{2}$ and therefore $P(A_2) = \frac{1}{2}$.

$\Rightarrow$ P(Getting a Head) $= \frac{1}{2}$ = P(Getting a Tail).

Thus, a coin is called unbiased, if the probability of getting head is equal to that of getting tail.

Aliter: (Applying Classical approach)

Since $n(A_1) = 1 = n(A_2)$ and $n(S) = 2$,

$$P(A_1) = \frac{n(A_1)}{n(S)} = \frac{1}{2}.$$

Similarly, $P(A_2) = \frac{1}{2}$.

Remark:

If a biased coin is tossed and the outcome of head is thrice as likely as tail, then $P(A_1) = 3P(A_2)$.

Substituting this in $P(S) = P(A_1)+P(A_2)$, it follows that

$$1 = P(A_1)+P(A_2) = 4P(A_2)$$

It gives that $P(A_2) = \frac{1}{4}$ and hence $P(A_1) = \frac{3}{4}$.

It should be noted that the probabilities for getting head and tail differ for a biased coin.